HW 4

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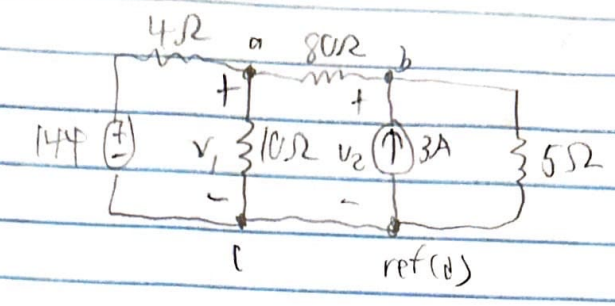
4.14:



a: $0 = \frac{V_a - 144}{4} + \frac{V_a}{10} + \frac{V_a - V_b}{80}$

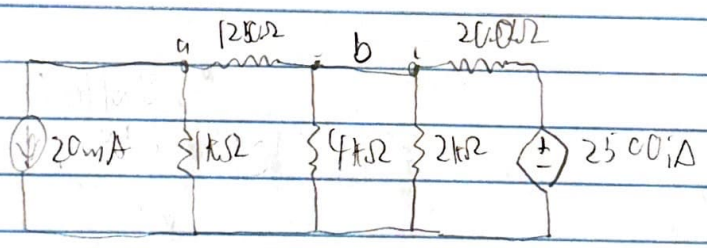
b: $0 = -3 + \frac{V_b - V_a}{80} + \frac{V_b}{50}$

done in matlab

V1 = 100 V

V2 = 20 V

4.19



$0 = .020 + \frac{V_a}{1k} + \frac{V_a - V_b}{1250}$

$\frac{V_b - V_a}{1250} = iΔ$

$0 = \frac{V_b}{4k} + \frac{V_b - V_a}{1250} + \frac{V_b}{2k} + \frac{V_b - 2500iΔ}{200}$

on matlab

$V_a = 60V$

$V_b = 160V$

$iΔ = .08A$

$V_a = 16{,}800$

$V_b = 12.8$

a. 1.2W absorbed

40W developed

$.02(60) = 1.2W$

$.08(2500) = 200V$

$\frac{160 - 200}{200} = -.2A$

$-.2(200) = -40W$

$.235 mW$

b. $\frac{60^2}{1000} = 3.6$

+

$\frac{160^2}{4000} = 6.4$

+

$1250(.08)(.08) = 8$

+

$\frac{160^2}{2000} = 12.8$

+

$\frac{40^2}{200} = 8$

40W

4.24

$$0 = 7 + \frac{V_0}{3} + \frac{V_0 - V_x}{1}$$

$$0 = -2V_x + \frac{V_x - V_0}{1} + \frac{V_x}{2}$$

$V_b = 1.5V$

$V_c = 9V$

$V_x = 5$

V_b is V_0

solved in matlab

$$\boxed{V_0 = 1.5V}$$

4.29

a: $0 = \frac{V_a - V_b}{2} + \frac{V_a}{20} + 35i$ thing

$0 = \frac{V_d - V_c}{4} + \frac{V_d}{80} + 3.125V$ 35i thing

and sup node

ignore

$$0 = \frac{V_a - 20}{2} + \frac{V_a}{20} + \frac{V_c - V_b}{4} + \frac{V_c}{80} + 3.125V_\Delta$$

$a - c = 35i_\phi$

$$0 = \frac{V_b}{40} + \frac{V_b - 20}{1} + \frac{V_b - V_c}{4}$$

$20 - V_b = V_\Delta$

$i_\phi = V_b / 40$

$V_a = -20.25V$

$V_b = 10V$

$V_c = -29V$

$i_\phi = .25A$

$V_\Delta = 10V$

$$\frac{-20.25 - 20}{2} + \frac{(10 - 20)}{1} = -30.125 \cdot 20 = -602.5W$$

$\boxed{602.5W \text{ produced}}$

4.4

$30 = -3i_2 + 30i_1 - 20i_3$

$0 = 53i_\Delta + 8i_2 - 3i_1 - 5i_3$

$30 = 27i_3 - 5i_2 - 20i_1$

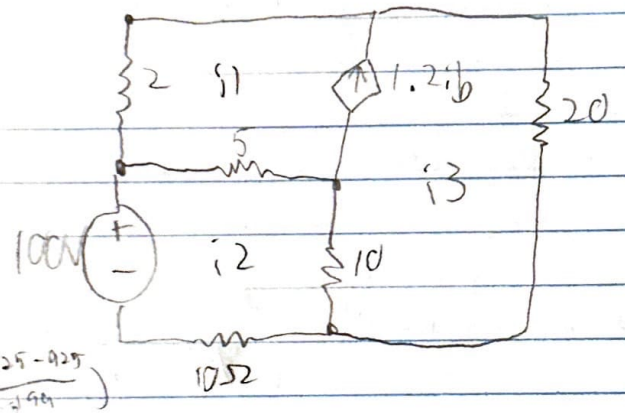
$i_\Delta = i_1 - i_3$

$i_1 = 52A \quad i_2 = 110A \quad i_3 = 60A$

$i_\Delta = -8A$

$-8(53)(110) = -46640\,W$

4.52



Power

$P_{dev} = -100(\frac{925}{199})$ $-232.65\,W$

$P_{dev} = -697.47\,W$ ✓

$P_a = 13.92\,W$

$P_b = 216.06\,W$

$P_c = 172.837\,W$

$P_d = 265.46\,W$

$P_e = 29.19\,W$

$P_{total} = 697.47\,W$ ✓

V_{cs}

$-2(-\frac{525}{199}) - 5(\frac{-525-925}{199})$

$V = 41.71$

$41.71(1.2(\frac{925}{199}))$

-232.65

$0 = 2i_1 + 20i_3 + 5(i_1 - i_2) + 10(i_3 - i_2)$

$0 = -100 + 5(i_2 - i_1) + 10i_2 + (i_2 - i_3)10$

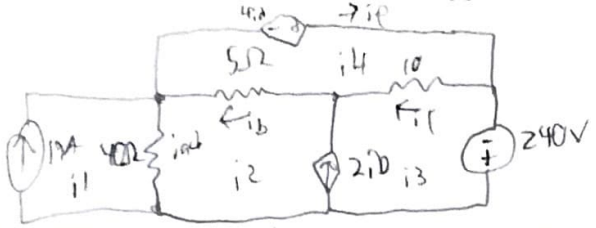
$i_3 - i_1 = 1.2i_b = 1.2i_2$

$i_1 = \frac{-525}{199} \quad i_2 = \frac{925}{199} \quad i_3 = \frac{585}{199}$

(in Amps)

$i_a = i_1 = \frac{-525}{199}, \quad i_b = i_2 = \frac{925}{199}, \quad i_c = i_3 = \frac{585}{199}, \quad i_d = i_2 - i_1 = \frac{1450}{199}$

$i_e = i_2 - i_3 = \frac{340}{199}$



4.53

$0 = i3 - i2 - 2(i4 - i2)$ $\quad i1 = 19$

$0 = (i2 - i1)40 + (i2 - i4)5 + v_{temp}$

$0 = (i2 - i1)40 + (i2 - i4)5 - 240 + (i3 - i4)10$

$0 = -240 + (i3 - i4)10 + v_{temp}$

$0 = (i4 - i3)10 + (i4 - i2)5 - 4i_d$

$i_b = i2(3)$

$i2 - i3 = i2(10)$

In matlab:

$i2 = 26A$, $i3 = 10A$, $i4 = 18A$

$i_a = 19 - 26 = -7A$

$i_b = 26 - 18 = 8A$

$i_c = 10 - 18 = -8A$

$i_d = 10A$

$i_e = 18A$

Power gen: $5120 + 720 + 2400 = 8240W$

Power disp: $5320 + 1960 + 320 + 640 = 8240W$

4.58

node voltage b/c I like it more

Supnode 1: $-.2 + \frac{V_b}{250} + \frac{V_a}{100} + .003V_\Delta = 0$

$V_b - V_a = 20$

Supnode 2: $.2 + \frac{V_c}{500} + \frac{V_d}{200} - .003V_\Delta = 0$

$V_d - V_c = .4V_{alpha}$

$V_b = V_{alpha}$

$V_c = V_\Delta$

$V_a = 24V$

$V_b = 44V$

$V_c = -72V$ $\quad \frac{44}{250} = .176$

$V_d = 54.4V$

$V_\Delta = -72V$ $\quad .2 - .176 = .024$

$V_{alpha} = 44V$ $\quad .024 * 20 = \boxed{.48W}$